

Tactile Intelligence Sharing-SSD2D0X 4G-cat1 Module Porting (I610)

原创

Industio_Haptic

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This article is based on SSD2D0X to transplant Guanghai Tong 4G-cat1 module.

Sigmastar Chip Platform-IDO-SOM2D0X



Ultra-small SOM module based on SigmaStar SSD201/202 SoC, dual-core A7 1.2GHz frequency, 1080P video **decoding**, support MIPI/RGB display interface, support dual Ethernet, support SDIO/USB/SPI/I2C/UART/DMIC/IIS, integrated audio Codec, supports microphone array; module integrates WIFI and NAND; supports Linux system; high performance, low cost; batch supply guarantee.

feature of product:

- Dual-core A7+Linux system, stable and smooth;
- MIPI/RGB display interface supports up to 1080P;
- Integrated DDR and PHY, support dual network ports, rich interfaces;
- 29.5*29.5mm ultra-small stamp hole size;

1. It can be seen from the schematic diagram that the 4G module uses GPIO39 GPIO40 GPIO41:



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The module uses the usb connection, we have already configured the USB, we just need to open the corresponding GPIO.

```
1 #####hub_usb4_4g_power_h#####
2 echo 39 > /sys/class/gpio/export
3 echo out > /sys/class/gpio/gpio39/direction
4 echo 1 > /sys/class/gpio/gpio39/value
5 #####hub_usb4_4g_power_on#####
6 echo 40 > /sys/class/gpio/export
7 echo out > /sys/class/gpio/gpio40/direction
8 echo 1 > /sys/class/gpio/gpio40/value
9 #####hub_usb4_4g_reset_N#####
10 echo 41 > /sys/class/gpio/export
11 echo out > /sys/class/gpio/gpio41/direction
12 echo 0 > /sys/class/gpio/gpio41/value
13 sleep 0.1
14 echo 1 > /sys/class/gpio/gpio41/value
```

2. Transplant 4G driver

Modify driver source code

```
# vi kernel/drivers/usb/serial/option.c
```

```
/* fibocom product */
#define FIBOCOM_VENDOR_ID          0x1782
#define FIBOCOM_PRODUCT_L61031    0x4D10
#define FIBOCOM_PRODUCT_L61032    0x4D11
#define FIBOCOM_PRODUCT_L61033    0x4D11
```

```
{ USB_DEVICE(FIBOCOM_VENDOR_ID,FIBOCOM_PRODUCT_L61031)},/*fibocom l610*/
{ USB_DEVICE(FIBOCOM_VENDOR_ID,FIBOCOM_PRODUCT_L61032)},
{ USB_DEVICE(FIBOCOM_VENDOR_ID,FIBOCOM_PRODUCT_L61033)},
{ } /* Terminating entry */
```

3. Add kernel configuration

CONFIG_USB_SERIAL=y

CONFIG_USB_SERIAL_OPTION=y

```
<*> USB driver for GSM and CDMA modems
```

4. Isusb can see and identify the module

```
# lsusb
Bus 002 Device 002: ID 1b20:8888
Bus 001 Device 001: ID 1d6b:0002
Bus 001 Device 002: ID 1a40:0101
Bus 002 Device 001: ID 1d6b:0002
Bus 001 Device 004: ID 1782:4d10
```

After power on, you can see the following print.

```

==20180309==> hub_port_init 4 #0
usb 1-1.4: new high-speed USB device number 3 using Sstar-ehci-2
usb 1-1.4: New USB device found, idVendor=1782, idProduct=4d12
usb 1-1.4: New USB device strings: Mfr=1, Product=2, SerialNumber=0
usb 1-1.4: Product: L610
usb 1-1.4: Manufacturer: FIBOCOM
usb 1-1.4: USB disconnect, device number 3
==20180309==> hub_port_init 4 #0
usb 1-1.4: new high-speed USB device number 4 using Sstar-ehci-2
usb 1-1.4: New USB device found, idVendor=1782, idProduct=4d10
usb 1-1.4: New USB device strings: Mfr=1, Product=2, SerialNumber=0
usb 1-1.4: Product: L610
usb 1-1.4: Manufacturer: FIBOCOM
option 1-1.4:1.0: GSM modem (1-port) converter detected
usb 1-1.4: GSM modem (1-port) converter now attached to ttyUSB0
option 1-1.4:1.1: GSM modem (1-port) converter detected
usb 1-1.4: GSM modem (1-port) converter now attached to ttyUSB1
option 1-1.4:1.2: GSM modem (1-port) converter detected
usb 1-1.4: GSM modem (1-port) converter now attached to ttyUSB2
option 1-1.4:1.3: GSM modem (1-port) converter detected
usb 1-1.4: GSM modem (1-port) converter now attached to ttyUSB3
option 1-1.4:1.4: GSM modem (1-port) converter detected
usb 1-1.4: GSM modem (1-port) converter now attached to ttyUSB4
option 1-1.4:1.5: GSM modem (1-port) converter detected
usb 1-1.4: GSM modem (1-port) converter now attached to ttyUSB5
option 1-1.4:1.6: GSM modem (1-port) converter detected
usb 1-1.4: GSM modem (1-port) converter now attached to ttyUSB6

```

5. Configure ppp

CONFIG_PPP=y

CONFIG_PPP_MULTILINK=y

CONFIG_PPP_FILTER=y

CONFIG_PPP_ASYNC=y

CONFIG_PPP_SYNC_TTY=y

CONFIG_PPP_DEFLATE=y

CONFIG_PPP_BSDCOMP=y

```

--- PPP (point-to-point protocol) support
<*>  PPP BSD-Compress compression
<*>  PPP Deflate compression
[*]   PPP filtering
<*>  PPP MPPE compression (encryption)
[*]   PPP multilink support
<*>  PPP over Ethernet
<*>  PPP support for async serial ports
<*>  PPP support for sync tty ports

```

6. Edit the dial-up script

```
# vi /etc/ppp/peers/modem_L610
```

```

1  nodetach
2  lock
3  /dev/ttyUSB0
4  115200
5  nocrtscts
6  debug
7  logfile /tmp/logfile
8  modem
9  hide-password
10 usepeerdns
11 noauth
12 noipdefault
13 novj
14 novjccomp
15 nocc
16 defaultroute
17 ipcp-accept-local
18 ipcp-accept-remote
19 connect '/usr/sbin/chat -s

```

```
disconnect '/usr/sbin/chat -s -v -f /etc/ppp/peers/chat-modem-disconnect'
```

```
# vi /etc/ppp/peers/chat-modem-connect
```

```
1 ABORT 'NOCARRIER'
2 ABORT 'ERROR'
3 ABORT 'NODIALTONE'
4 ABORT 'BUSY'
5 ABORT 'NOANSWER'
6 '' AT
7 OK AT+GTRAT?
8 OK AT+CMEE=2
9 OK AT+CSQ
10 OK AT+CPIN?
11 OK AT+COPS?
12 OK AT+CGDCONT=1,\"IP\",,\"cmnet\"
13 OK ATDT*99#
14 CONNECT''
```

```
# vi /etc/ppp/peers/chat-modem-disconnect
```

7. ppp dial

pppd call modem_L610 &

```
Terminating on signal 2
Connect time 0.3 minutes.
Sent 0 bytes, received 0 bytes.
sent [LCP TermReq id=0x3 "user request"]
rcvd [LCP TermAck id=0x3]
Connection terminated.
abort on (OK)
abort on (BUSY)
abort on (DELAYED)
abort on (NO ANSWER)
abort on (NO CARRIER)
abort on (NO DIALTONE)
abort on (VOICE)
abort on (ERROR)
abort on (RINGING)
timeout set to 12 seconds
send (\K^M)
send (\K^M)
send (\K^M)
send (++ATH^M)
send (++ATH^M)
send (++ATH^M)
SIGINT
child process chat -s -v -f /etc/ppp/peers/chat-modem-disconnect (pid 1167) terminated with signal 2
disconnect script failed
abort on (NO CARRIER)
abort on (ERROR)
abort on (NO DIALTONE)
abort on (BUSY)
abort on (NO ANSWER)
send (AT^M)
expect (OK)
AT^M^M
OK
-- got it

send (AT+GTRAT?^M)
expect (OK)
^M
AT+GTRAT?^M^M
+GTRAT: 10,3,0^M
^M
OK
-- got it

send (AT+CMEE=2^M)
expect (OK)
^M
```

Try to ping 114.114.114.114 to verify that the network is working.

Connected successfully!

The Future of Tactile Sensing- Shenzhen Tactile Intelligence Technology Co., Ltd. (industio.cn)

For more product details, please search and follow the WX official account: Shenzhen Tactile Intelligent Technology Co., Ltd., more dry goods are waiting for you!

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